

СОДЕРЖАНИЕ

ЗАДАНИЕ 1	2
ЗАДАНИЕ 2	40
ЗАДАНИЕ 3	44
ЗАДАНИЕ 4	46

ЗАДАНИЕ 1

1) Для заданной на схеме schema-lab3 сети, состоящей из управляемых коммутаторов и персональных компьютеров

настроить на коммутаторах логическую топологию используя протокол IEEE 802.1Q, для передачи пакетов VLAN333 между коммутаторами использовать Native VLAN

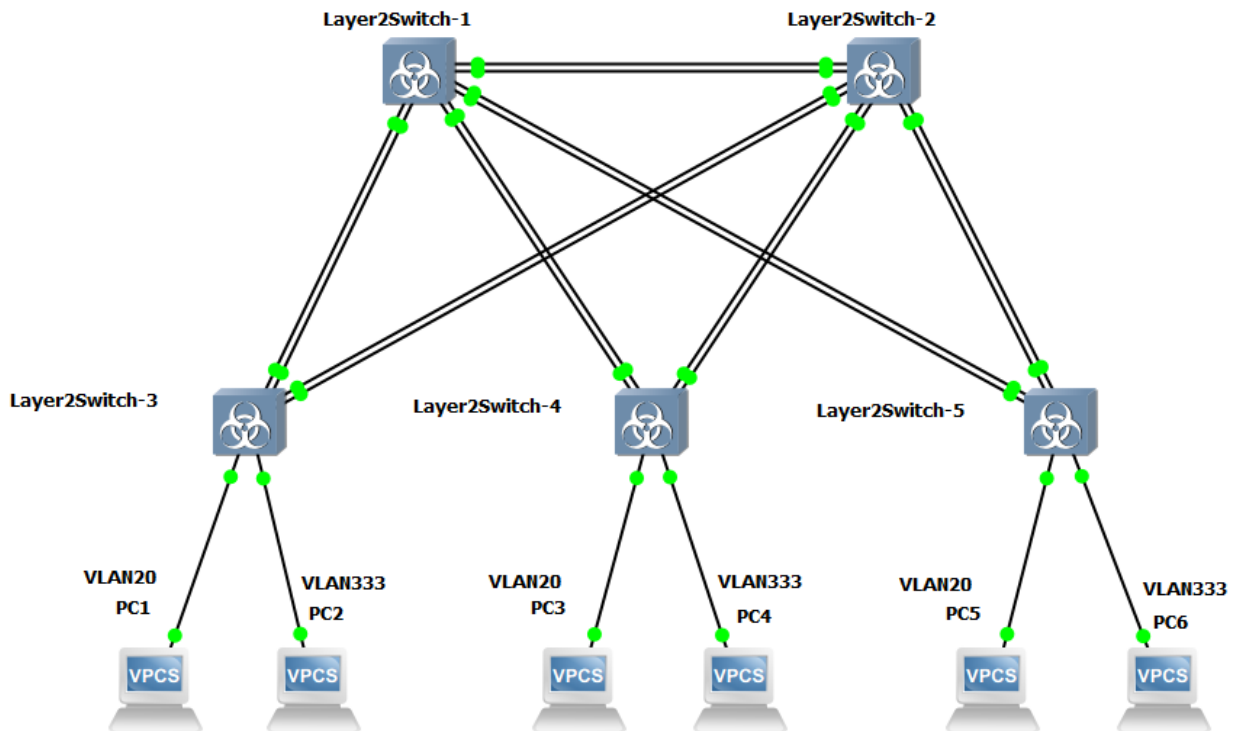


Рисунок 1 — Физическая топология сети для настройки VLAN

```
Switch>enable
```

```
Switch#show cdp neighbors
```

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge

S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,

D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
Switch	Gig 1/0	168	R S I IOSv	Gig 0/0	
Switch	Gig 1/1	162	R S I IOSv	Gig 0/1	
Switch	Gig 1/3	172	R S I IOSv	Gig 0/1	

Switch	Gig 1/2	159	R S I IOSv	Gig 0/0
Switch	Gig 0/2	175	R S I IOSv	Gig 0/0
Switch	Gig 0/3	179	R S I IOSv	Gig 0/1
Switch	Gig 0/0	136	R S I IOSv	Gig 0/0
Switch	Gig 0/1	141	R S I IOSv	Gig 0/1

Total cdp entries displayed : 8

Switch>enable

Switch#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#hostname Layer2Switch-2

Layer2Switch-2(config)#end

Layer2Switch-2#

Switch>enable

Switch#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#hostname Layer2Switch-3

Layer2Switch-3(config)#end

Layer2Switch-3#

Switch>enable

Switch#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#hostname Layer2Switch-4

Layer2Switch-4(config)#end

Layer2Switch-4#

Switch>enable

Switch#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#hostname Layer2Switch-5

Layer2Switch-5(config)#end

Layer2Switch-5#

Layer2Switch-1>show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge

S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,

D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
Switch	Gig 1/0	71	R S I IOSv	Gig 0/0	
Switch	Gig 1/1	83	R S I IOSv	Gig 0/1	
Switch	Gig 1/3	100	R S I IOSv	Gig 0/1	
Switch	Gig 1/2	97	R S I IOSv	Gig 0/0	
Switch	Gig 0/2	48	R S I IOSv	Gig 0/0	
Switch	Gig 0/3	56	R S I IOSv	Gig 0/1	
Switch	Gig 0/0	19	R S I IOSv	Gig 0/0	
Layer2Switch-4	Gig 1/1	172	R S I IOSv	Gig 0/1	
Layer2Switch-4	Gig 1/0	163	R S I IOSv	Gig 0/0	
Layer2Switch-5	Gig 1/3	143	R S I IOSv	Gig 0/1	
Layer2Switch-5	Gig 1/2	143	R S I IOSv	Gig 0/0	
Layer2Switch-2	Gig 0/0	128	R S I IOSv	Gig 0/0	
Layer2Switch-2	Gig 0/1	157	R S I IOSv	Gig 0/1	
Layer2Switch-3	Gig 0/3	146	R S I IOSv	Gig 0/1	

Layer2Switch-3 Gig 0/2 178 R S I IOSv Gig 0/0

Total cdp entries displayed : 15

Layer2Switch-1>enable

Layer2Switch-1#clear cdp table

Layer2Switch-1#show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge

S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,

D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
Layer2Switch-4	Gig 1/1	169	R S I IOSv		Gig 0/1
Layer2Switch-4	Gig 1/0	165	R S I IOSv		Gig 0/0
Layer2Switch-5	Gig 1/3	151	R S I IOSv		Gig 0/1
Layer2Switch-5	Gig 1/2	141	R S I IOSv		Gig 0/0
Layer2Switch-2	Gig 0/0	163	R S I IOSv		Gig 0/0
Layer2Switch-2	Gig 0/1	146	R S I IOSv		Gig 0/1
Layer2Switch-3	Gig 0/3	156	R S I IOSv		Gig 0/1
Layer2Switch-3	Gig 0/2	176	R S I IOSv		Gig 0/0

Total cdp entries displayed : 8

Layer2Switch-1#enable

Layer2Switch-1#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Layer2Switch-1(config)#vlan 20

```
Layer2Switch-1(config-vlan)#name VLAN20
Layer2Switch-1(config-vlan)#exit
Layer2Switch-1(config)#vlan 333
Layer2Switch-1(config-vlan)#name VLAN333
Layer2Switch-1(config-vlan)#exit
```

```
Layer2Switch-2#enable
Layer2Switch-2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Layer2Switch-2(config)#vlan 20
Layer2Switch-2(config-vlan)#name VLAN20
Layer2Switch-2(config-vlan)#exit
Layer2Switch-2(config)#vlan 333
Layer2Switch-2(config-vlan)#name VLAN333
Layer2Switch-2(config-vlan)#exit
Layer2Switch-2(config)#end
Layer2Switch-2#
```

```
Layer2Switch-3#enable
Layer2Switch-3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Layer2Switch-3(config)#vlan 20
Layer2Switch-3(config-vlan)#name VLAN20
Layer2Switch-3(config-vlan)#exit
Layer2Switch-3(config)#vlan 333
Layer2Switch-3(config-vlan)#name VLAN333
Layer2Switch-3(config-vlan)#exit
Layer2Switch-3(config)#end
```

Layer2Switch-3#

Layer2Switch-4#enable

Layer2Switch-4#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Layer2Switch-4(config)#vlan 20

Layer2Switch-4(config-vlan)#name VLAN20

Layer2Switch-4(config-vlan)#exit

Layer2Switch-4(config)#vlan 333

Layer2Switch-4(config-vlan)#name VLAN333

Layer2Switch-4(config-vlan)#exit

Layer2Switch-4(config)#end

Layer2Switch-4#

Layer2Switch-5>enable

Layer2Switch-5#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Layer2Switch-5(config)#vlan 20

Layer2Switch-5(config-vlan)#name VLAN20

Layer2Switch-5(config-vlan)#exit

Layer2Switch-5(config)#vlan 333

Layer2Switch-5(config-vlan)#name VLAN333

Layer2Switch-5(config-vlan)#exit

Layer2Switch-5(config)#end

Layer2Switch-5#

Layer2Switch-1>show vlan brief

VLAN Name	Status	Ports

1 default	active	Gi2/0
20 VLAN20	active	
100 VLAN100	active	
200 VLAN0200	active	
300 VLAN0300	active	
333 VLAN333	active	
1002 fddi-default	act/unsup	
1003 trcrf-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trbrf-default	act/unsup	
Layer2Switch-1>		

Layer2Switch-3>enable

Layer2Switch-3#show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
Layer2Switch-2	Gig 0/2	127	R S I IOSv		Gig 0/2
Layer2Switch-2	Gig 0/3	137	R S I IOSv		Gig 0/3
Layer2Switch-1	Gig 0/0	118	R S I IOSv		Gig 0/2
Layer2Switch-1	Gig 0/1	169	R S I IOSv		Gig 0/3

Total cdp entries displayed : 4

Layer2Switch-3#show interfaces status

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/0		connected	trunk	auto	auto	unknown
Gi0/1		connected	trunk	auto	auto	unknown
Gi0/2		connected	trunk	auto	auto	unknown
Gi0/3		connected	trunk	auto	auto	unknown
Gi1/0		connected	unassigned	auto	auto	unknown
Gi1/1		connected	unassigned	auto	auto	unknown

Layer2Switch-3#

Layer2Switch-4>enable

Layer2Switch-4#show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
 S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
 D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
Layer2Switch-2	Gig 0/3	144	R S I IOSv		Gig 1/1
Layer2Switch-2	Gig 0/2	173	R S I IOSv		Gig 1/0
Layer2Switch-1	Gig 0/0	115	R S I IOSv		Gig 1/0
Layer2Switch-1	Gig 0/1	156	R S I IOSv		Gig 1/1

Total cdp entries displayed : 4

Layer2Switch-4#show interfaces status

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/0		connected	trunk	auto	auto	unknown
Gi0/1		connected	trunk	auto	auto	unknown
Gi0/2		connected	trunk	auto	auto	unknown

```

Gi0/3          connected trunk    auto  auto unknown
Gi1/0          connected unassigned auto  auto unknown
Gi1/1          connected unassigned auto  auto unknown
Layer2Switch-4#

```

Layer2Switch-5#enable

Layer2Switch-5#show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
 S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
 D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
Layer2Switch-2	Gig 0/3	148	R S I IOSv		Gig 1/3
Layer2Switch-2	Gig 0/2	162	R S I IOSv		Gig 1/2
Layer2Switch-1	Gig 0/0	173	R S I IOSv		Gig 1/2
Layer2Switch-1	Gig 0/1	136	R S I IOSv		Gig 1/3

Total cdp entries displayed : 4

Layer2Switch-5#show interfaces status

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/0		connected	trunk	auto	auto	unknown
Gi0/1		connected	trunk	auto	auto	unknown
Gi0/2		connected	trunk	auto	auto	unknown
Gi0/3		connected	trunk	auto	auto	unknown
Gi1/0		connected	unassigned	auto	auto	unknown
Gi1/1		connected	unassigned	auto	auto	unknown

Layer2Switch-5#

Layer2Switch-3#enable

Layer2Switch-3#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Layer2Switch-3(config)#interface GigabitEthernet1/0

Layer2Switch-3(config-if)#switchport mode access

Layer2Switch-3(config-if)#switchport access vlan 20

Layer2Switch-3(config-if)#

*Sep 8 12:19:29.950: %PM-3-INTERNALERROR: Port Manager Internal Software Error (vlan > 0 && vlan < PM_MAX_VLANS: ../switch/pm/pm_vlan.c: 870: pm_vlan_test_portlist)

-Traceback= 1C492C9z F385ABz FA54E0z FA5617z F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290D9Bz 3290B5Cz 3291A3Az

*Sep 8 12:19:29.953: %PM-3-INTERNALERROR: Port Manager Internal Software Error (pm_vlan_test_portlist(vlan, pd->globalNumber): ../switch/pm/pm_vlan.c: 1777: pm_vlan_rem_port)

-Traceback= 1C492C9z F385ABz FA578Bz F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290D9Bz 3290B5Cz 3291A3Az F67FF2z

*Sep 8 12:19:29.953: %PM-3-INTERNALERROR: Port Manager Internal Software Error (vlan > 0 && vlan < PM_MAX_VLANS: ../switch/pm/pm_vlan.c: 860: pm_vlan_clear_portlist)

-Traceback= 1C492C9z F385ABz FA58F1z FA5665z F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290D9Bz 3290B5Cz 3291A3Az

*Sep 8 12:19:29.954: %BIT-4-OUTOFRANGE: bit 0 is not in the expected range of 1 to 4095

-Traceback= 1C492C9z 196C71Fz 196BF98z FA5699z F9398Bz F93DD7z
3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z
3290D9Bz 3290B5Cz 3291A3Az

*Sep 8 12:19:29.955: %PM-3-INTERNALERROR: Port Manager Internal
Software Error (pm_vtpvlan_bitlist_test(&pd->operInfo.trunkVlans, vlan):
../switch/pm/pm_vlan.c: 1789: pm_vlan_rem_port)

-Traceback= 1C492C9z F385ABz FA5700z F9398Bz F93DD7z 3013F8Fz
F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290D9Bz
3290B5Cz 3291A3Az F67FF2z

*Sep 8 12:19:29.957: %BIT-4-OUTOFRANGE: bit 0 is not in the expected
range of 1 to 4095

-Traceback= 1C492C9z 196C71Fz 196C008z FA56A7z F9398Bz F93DD7z
3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z
3290D9Bz 3290B5Cz 3291A3Az

Layer2Switch-3(config-if)#

Layer2Switch-3(config-if)#exit

Layer2Switch-3(config)#interface GigabitEthernet1/1

Layer2Switch-3(config-if)#switchport mode access

Layer2Switch-3(config-if)#switchport access vlan 333

Layer2Switch-3(config-if)#

*Sep 8 12:19:56.743: %PM-3-INTERNALERROR: Port Manager Internal
Software Error (vlan > 0 && vlan < PM_MAX_VLANS: ../switch/pm/pm_vlan.c:
870: pm_vlan_test_portlist)

-Traceback= 1C492C9z F385ABz FA54E0z FA5617z F9398Bz F93DD7z
3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z
3290D9Bz 3290B5Cz 3291A3Az

*Sep 8 12:19:56.743: %PM-3-INTERNALERROR: Port Manager Internal
Software Error (pm_vlan_test_portlist(vlan, pd->globalNumber):
../switch/pm/pm_vlan.c: 1777: pm_vlan_rem_port)

-Traceback= 1C492C9z F385ABz FA578Bz F9398Bz F93DD7z 3013F8Fz
F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290D9Bz
3290B5Cz 3291A3Az F67FF2z

*Sep 8 12:19:56.745: %PM-3-INTERNALERROR: Port Manager Internal
Software Error (vlan > 0 && vlan < PM_MAX_VLANS: ../switch/pm/pm_vlan.c:
860: pm_vlan_clear_portlist)

-Traceback= 1C492C9z F385ABz FA58F1z FA5665z F9398Bz F93DD7z
3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z
3290D9Bz 3290B5Cz 3291A3Az

*Sep 8 12:19:56.745: %BIT-4-OUTOFRANGE: bit 0 is not in the expected
range of 1 to 4095

-Traceback= 1C492C9z 196C71Fz 196BF98z FA5699z F9398Bz F93DD7z
3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z
3290D9Bz 3290B5Cz 3291A3Az

*Sep 8 12:19:56.745: %PM-3-INTERNALERROR: Port Manager Internal
Software Error (pm_vtpvlan_bitlist_test(&pd->operInfo.trunkVlans, vlan):
../switch/pm/pm_vlan.c: 1789: pm_vlan_rem_port)

-Traceback= 1C492C9z F385ABz FA5700z F9398Bz F93DD7z 3013F8Fz
F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290D9Bz
3290B5Cz 3291A3Az F67FF2z

*Sep 8 12:19:56.745: %BIT-4-OUTOFRANGE: bit 0 is not in the expected
range of 1 to 4095

-Traceback= 1C492C9z 196C71Fz 196C008z FA56A7z F9398Bz F93DD7z
3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z
3290D9Bz 3290B5Cz 3291A3Az

Layer2Switch-3(config-if)#exit

Layer2Switch-3(config)#end

Layer2Switch-3#

*Sep 8 12:20:14.015: %SYS-5-CONFIG_I: Configured from console by
console

Layer2Switch-3#

Layer2Switch-4#enable

Layer2Switch-4#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Layer2Switch-4(config)#interface GigabitEthernet1/0

Layer2Switch-4(config-if)#switchport mode access

Layer2Switch-4(config-if)#switchport access vlan 20

Layer2Switch-4(config-if)#

*Sep 8 12:22:41.269: %PM-3-INTERNALERROR: Port Manager Internal Software Error (vlan > 0 && vlan < PM_MAX_VLANS: ../switch/pm/pm_vlan.c: 870: pm_vlan_test_portlist)

-Traceback= 1C492C9z F385ABz FA54E0z FA5617z F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290D9Bz 3290B5Cz 3291A3Az

*Sep 8 12:22:41.279: %PM-3-INTERNALERROR: Port Manager Internal Software Error (pm_vlan_test_portlist(vlan, pd->globalNumber): ../switch/pm/pm_vlan.c: 1777: pm_vlan_rem_port)

-Traceback= 1C492C9z F385ABz FA578Bz F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290D9Bz 3290B5Cz 3291A3Az F67FF2z

*Sep 8 12:22:41.279: %PM-3-INTERNALERROR: Port Manager Internal Software Error (vlan > 0 && vlan < PM_MAX_VLANS: ../switch/pm/pm_vlan.c: 860: pm_vlan_clear_portlist)

-Traceback= 1C492C9z F385ABz FA58F1z FA5665z F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290D9Bz 3290B5Cz 3291A3Az

*Sep 8 12:22:41.281: %BIT-4-OUTOFRANGE: bit 0 is not in the expected range of 1 to 4095

-Traceback= 1C492C9z 196C71Fz 196BF98z FA5699z F9398Bz F93DD7z
3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z
3290D9Bz 3290B5Cz 3291A3Az

*Sep 8 12:22:41.282: %PM-3-INTERNALERROR: Port Manager Internal
Software Error (pm_vtpvlan_bitlist_test(&pd->operInfo.trunkVlans, vlan):
../switch/pm/pm_vlan.c: 1789: pm_vlan_rem_port)

-Traceback= 1C492C9z F385ABz FA5700z F9398Bz F93DD7z 3013F8Fz
F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290D9Bz
3290B5Cz 3291A3Az F67FF2z

*Sep 8 12:22:41.282: %BIT-4-OUTOFRANGE: bit 0 is not in the expected
range of 1 to 4095

-Traceback= 1C492C9z 196C71Fz 196C008z FA56A7z F9398Bz F93DD7z
3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z
3290D9Bz 3290B5Cz 3291A3Az

Layer2Switch-4(config-if)#exit

Layer2Switch-4(config)#interface GigabitEthernet1/1

Layer2Switch-4(config-if)#switchport mode access

Layer2Switch-4(config-if)#switchport access vlan 333

Layer2Switch-4(config-if)#

*Sep 8 12:22:55.445: %PM-3-INTERNALERROR: Port Manager Internal
Software Error (vlan > 0 && vlan < PM_MAX_VLANS: ../switch/pm/pm_vlan.c:
870: pm_vlan_test_portlist)

-Traceback= 1C492C9z F385ABz FA54E0z FA5617z F9398Bz F93DD7z
3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z
3290D9Bz 3290B5Cz 3291A3Az

*Sep 8 12:22:55.445: %PM-3-INTERNALERROR: Port Manager Internal
Software Error (pm_vlan_test_portlist(vlan, pd->globalNumber):
../switch/pm/pm_vlan.c: 1777: pm_vlan_rem_port)

-Traceback= 1C492C9z F385ABz FA578Bz F9398Bz F93DD7z 3013F8Fz
F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290D9Bz
3290B5Cz 3291A3Az F67FF2z

*Sep 8 12:22:55.446: %PM-3-INTERNALERROR: Port Manager Internal
Software Error (vlan > 0 && vlan < PM_MAX_VLANS: ../switch/pm/pm_vlan.c:
860: pm_vlan_clear_portlist)

-Traceback= 1C492C9z F385ABz FA58F1z FA5665z F9398Bz F93DD7z
3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z
3290D9Bz 3290B5Cz 3291A3Az

*Sep 8 12:22:55.447: %BIT-4-OUTOFRANGE: bit 0 is not in the expected
range of 1 to 4095

-Traceback= 1C492C9z 196C71Fz 196BF98z FA5699z F9398Bz F93DD7z
3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z
3290D9Bz 3290B5Cz 3291A3Az

*Sep 8 12:22:55.447: %PM-3-INTERNALERROR: Port Manager Internal
Software Error (pm_vtpvlan_bitlist_test(&pd->operInfo.trunkVlans, vlan):
../switch/pm/pm_vlan.c: 1789: pm_vlan_rem_port)

-Traceback= 1C492C9z F385ABz FA5700z F9398Bz F93DD7z 3013F8Fz
F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290D9Bz
3290B5Cz 3291A3Az F67FF2z

*Sep 8 12:22:55.447: %BIT-4-OUTOFRANGE: bit 0 is not in the expected
range of 1 to 4095

-Traceback= 1C492C9z 196C71Fz 196C008z FA56A7z F9398Bz F93DD7z
3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z
3290D9Bz 3290B5Cz 3291A3Az

Layer2Switch-4(config-if)#exit

Layer2Switch-4(config)#end

Layer2Switch-4#

*Sep 8 12:23:04.520: %SYS-5-CONFIG_I: Configured from console by
console

Layer2Switch-5#enable

Layer2Switch-5#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Layer2Switch-5(config)#interface GigabitEthernet1/0

Layer2Switch-5(config-if)#switchport mode access

Layer2Switch-5(config-if)#switchport access vlan 20

Layer2Switch-5(config-if)#

*Sep 8 12:20:44.614: %PM-3-INTERNALERROR: Port Manager Internal Software Error (vlan > 0 && vlan < PM_MAX_VLANS: ../switch/pm/pm_vlan.c: 870: pm_vlan_test_portlist)

-Traceback= 1C492C9z F385ABz FA54E0z FA5617z F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290D9Bz 3290B5Cz 3291A3Az

*Sep 8 12:20:44.617: %PM-3-INTERNALERROR: Port Manager Internal Software Error (pm_vlan_test_portlist(vlan, pd->globalNumber): ../switch/pm/pm_vlan.c: 1777: pm_vlan_rem_port)

-Traceback= 1C492C9z F385ABz FA578Bz F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290D9Bz 3290B5Cz 3291A3Az F67FF2z

*Sep 8 12:20:44.617: %PM-3-INTERNALERROR: Port Manager Internal Software Error (vlan > 0 && vlan < PM_MAX_VLANS: ../switch/pm/pm_vlan.c: 860: pm_vlan_clear_portlist)

-Traceback= 1C492C9z F385ABz FA58F1z FA5665z F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290D9Bz 3290B5Cz 3291A3Az

*Sep 8 12:20:44.618: %BIT-4-OUTOFRANGE: bit 0 is not in the expected range of 1 to 4095

-Traceback= 1C492C9z 196C71Fz 196BF98z FA5699z F9398Bz F93DD7z
3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z
3290D9Bz 3290B5Cz 3291A3Az

*Sep 8 12:20:44.619: %PM-3-INTERNALERROR: Port Manager Internal
Software Error (pm_vtpvlan_bitlist_test(&pd->operInfo.trunkVlans, vlan):
../switch/pm/pm_vlan.c: 1789: pm_vlan_rem_port)

-Traceback= 1C492C9z F385ABz FA5700z F9398Bz F93DD7z 3013F8Fz
F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290D9Bz
3290B5Cz 3291A3Az F67FF2z

*Sep 8 12:20:44.619: %BIT-4-OUTOFRANGE: bit 0 is not in the expected
range of 1 to 4095

-Traceback= 1C492C9z 196C71Fz 196C008z FA56A7z F9398Bz F93DD7z
3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z
3290D9Bz 3290B5Cz 3291A3Az

Layer2Switch-5(config-if)#exit

Layer2Switch-5(config)#interface GigabitEthernet1/1

Layer2Switch-5(config-if)#switchport mode access

Layer2Switch-5(config-if)#switchport access vlan 333

Layer2Switch-5(config-if)#

*Sep 8 12:21:04.346: %PM-3-INTERNALERROR: Port Manager Internal
Software Error (vlan > 0 && vlan < PM_MAX_VLANS: ../switch/pm/pm_vlan.c:
870: pm_vlan_test_portlist)

-Traceback= 1C492C9z F385ABz FA54E0z FA5617z F9398Bz F93DD7z
3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z
3290D9Bz 3290B5Cz 3291A3Az

*Sep 8 12:21:04.346: %PM-3-INTERNALERROR: Port Manager Internal
Software Error (pm_vlan_test_portlist(vlan, pd->globalNumber):
../switch/pm/pm_vlan.c: 1777: pm_vlan_rem_port)

-Traceback= 1C492C9z F385ABz FA578Bz F9398Bz F93DD7z 3013F8Fz
F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290D9Bz
3290B5Cz 3291A3Az F67FF2z

*Sep 8 12:21:04.347: %PM-3-INTERNALERROR: Port Manager Internal
Software Error (vlan > 0 && vlan < PM_MAX_VLANS: ../switch/pm/pm_vlan.c:
860: pm_vlan_clear_portlist)

-Traceback= 1C492C9z F385ABz FA58F1z FA5665z F9398Bz F93DD7z
3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z
3290D9Bz 3290B5Cz 3291A3Az

*Sep 8 12:21:04.348: %BIT-4-OUTOFRANGE: bit 0 is not in the expected
range of 1 to 4095

-Traceback= 1C492C9z 196C71Fz 196BF98z FA5699z F9398Bz F93DD7z
3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z
3290D9Bz 3290B5Cz 3291A3Az

*Sep 8 12:21:04.348: %PM-3-INTERNALERROR: Port Manager Internal
Software Error (pm_vtpvlan_bitlist_test(&pd->operInfo.trunkVlans, vlan):
../switch/pm/pm_vlan.c: 1789: pm_vlan_rem_port)

-Traceback= 1C492C9z F385ABz FA5700z F9398Bz F93DD7z 3013F8Fz
F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290D9Bz
3290B5Cz 3291A3Az F67FF2z

*Sep 8 12:21:04.348: %BIT-4-OUTOFRANGE: bit 0 is not in the expected
range of 1 to 4095

-Traceback= 1C492C9z 196C71Fz 196C008z FA56A7z F9398Bz F93DD7z
3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z
3290D9Bz 3290B5Cz 3291A3Az

Layer2Switch-5(config-if)#exit

Layer2Switch-5(config)#end

Layer2Switch-5#

*Sep 8 12:21:11.119: %SYS-5-CONFIG_I: Configured from console by
console

Layer2Switch-3#show vlan brief

VLAN Name	Status	Ports

1 default	active	
20 VLAN20	active	Gi1/0
100 VLAN100	active	
200 VLAN0200	active	
300 VLAN0300	active	
333 VLAN333	active	Gi1/1
1002 fddi-default	act/unsup	
1003 trcrf-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trbrf-default	act/unsup	

Layer2Switch-3#

Layer2Switch-4#show vlan brief

VLAN Name	Status	Ports

1 default	active	
20 VLAN20	active	Gi1/0
100 VLAN100	active	
200 VLAN0200	active	
300 VLAN0300	active	
333 VLAN333	active	Gi1/1
1002 fddi-default	act/unsup	

```

1003 trcrf-default          act/unsup
1004 fddinet-default        act/unsup
1005 trbrf-default          act/unsup
Layer2Switch-4#

```

Layer2Switch-5#show vlan brief

VLAN Name	Status	Ports

1 default	active	
20 VLAN20	active	Gi1/0
100 VLAN100	active	
200 VLAN0200	active	
300 VLAN0300	active	
333 VLAN333	active	Gi1/1
1002 fddi-default	act/unsup	
1003 trcrf-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trbrf-default	act/unsup	

Layer2Switch-5#

Layer2Switch-3#show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	1
Gi0/1	on	802.1q	trunking	333
Gi0/2	desirable	n-isl	trunking	1
Gi0/3	desirable	n-isl	trunking	1

Port	Vlans allowed on trunk
Gi0/0	20
Gi0/1	333
Gi0/2	1-4094
Gi0/3	1-4094

Port	Vlans allowed and active in management domain
Gi0/0	20
Gi0/1	333
Gi0/2	1,20,100,200,300,333
Gi0/3	1,20,100,200,300,333

Port	Vlans in spanning tree forwarding state and not pruned
Gi0/0	20
Gi0/1	333
Gi0/2	1,20,100,200,300,333

Port	Vlans in spanning tree forwarding state and not pruned
Gi0/3	20,100,200,300,333

Layer2Switch-3#show running-config | section interface

interface GigabitEthernet0/0

switchport trunk encapsulation dot1q

switchport trunk allowed vlan 20

switchport mode trunk

media-type rj45

negotiation auto

interface GigabitEthernet0/1

switchport trunk encapsulation dot1q

switchport trunk native vlan 333

```
switchport trunk allowed vlan 333
```

```
switchport mode trunk
```

```
media-type rj45
```

```
negotiation auto
```

```
interface GigabitEthernet0/2
```

```
media-type rj45
```

```
negotiation auto
```

```
interface GigabitEthernet0/3
```

```
media-type rj45
```

```
negotiation auto
```

```
interface GigabitEthernet1/0
```

```
switchport access vlan 20
```

```
media-type rj45
```

```
negotiation auto
```

```
interface GigabitEthernet1/1
```

```
switchport access vlan 333
```

```
media-type rj45
```

```
negotiation auto
```

```
Layer2Switch-3#
```

```
Layer2Switch-3#configure terminal
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
Layer2Switch-3(config)#interface GigabitEthernet0/2
```

```
Layer2Switch-3(config-if)#switchport trunk encapsulation dot1q
```

```
Layer2Switch-3(config-if)#switchport trunk allowed vlan 20
```

```
Layer2Switch-3(config-if)#switchport mode trunk
```

```
Layer2Switch-3(config-if)#exit
```

```
Layer2Switch-3(config)#interface GigabitEthernet0/3
```

```
Layer2Switch-3(config-if)#switchport trunk encapsulation dot1q
```

```
Layer2Switch-3(config-if)#switchport trunk native vlan 333
```

```
Layer2Switch-3(config-if)#switchport trunk allowed vlan 333
```

```
Layer2Switch-3(config-if)#switchport mode trunk
```

```
Layer2Switch-3(config-if)#exit
```

```
Layer2Switch-3(config)#end
```

```
Layer2Switch-3#
```

```
*Sep  8 12:27:06.741: %SYS-5-CONFIG_I: Configured from console by console
```

```
Layer2Switch-3#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	1
Gi0/1	on	802.1q	trunking	333
Gi0/2	on	802.1q	trunking	1
Gi0/3	on	802.1q	trunking	333

Port	Vlans allowed on trunk
Gi0/0	20
Gi0/1	333
Gi0/2	20
Gi0/3	333

Port	Vlans allowed and active in management domain
Gi0/0	20
Gi0/1	333
Gi0/2	20
Gi0/3	333

Port	Vlans in spanning tree forwarding state and not pruned
------	--

Gi0/0	20
Gi0/1	333
Gi0/2	none

Port	Vlans in spanning tree forwarding state and not pruned
Gi0/3	none

Layer2Switch-3#

Layer2Switch-1>show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	desirable	n-isl	trunking	1
Gi0/1	desirable	n-isl	trunking	1
Gi0/2	on	802.1q	trunking	1
Gi0/3	on	802.1q	trunking	333
Gi1/0	on	802.1q	trunking	1
Gi1/1	on	802.1q	trunking	333
Gi1/2	on	802.1q	trunking	1
Gi1/3	on	802.1q	trunking	333

Port	Vlans allowed on trunk
Gi0/0	1-4094
Gi0/1	1-4094
Gi0/2	20
Gi0/3	333
Gi1/0	20
Gi1/1	333
Gi1/2	20
Gi1/3	333

Port	Vlans allowed and active in management domain
Gi0/0	1,20,100,200,300,333

Port	Vlans allowed and active in management domain
Gi0/1	1,20,100,200,300,333
Gi0/2	20
Gi0/3	333
Gi1/0	20
Gi1/1	333
Gi1/2	20
Gi1/3	333

Port	Vlans in spanning tree forwarding state and not pruned
Gi0/0	1,20,100,200,300,333
Gi0/1	20,100,200,300,333
Gi0/2	20
Gi0/3	333
Gi1/0	20
Gi1/1	333
Gi1/2	20
Gi1/3	333

Layer2Switch-1>

Layer2Switch-2>show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	desirable	n-isl	trunking	1
Gi0/1	desirable	n-isl	trunking	1

Gi0/2	desirable	n-802.1q	trunking	1
Gi0/3	desirable	n-802.1q	trunking	1
Gi1/0	desirable	n-isl	trunking	1
Gi1/1	desirable	n-isl	trunking	1
Gi1/2	desirable	n-isl	trunking	1
Gi1/3	desirable	n-isl	trunking	1

Port Vlan allowed on trunk

Gi0/0	1-4094
Gi0/1	1-4094
Gi0/2	1-4094
Gi0/3	1-4094
Gi1/0	1-4094
Gi1/1	1-4094
Gi1/2	1-4094
Gi1/3	1-4094

Port Vlan allowed and active in management domain

Gi0/0	1,20,100,200,300,333
-------	----------------------

Port Vlan allowed and active in management domain

Gi0/1	1,20,100,200,300,333
Gi0/2	1,20,100,200,300,333
Gi0/3	1,20,100,200,300,333
Gi1/0	1,20,100,200,300,333
Gi1/1	1,20,100,200,300,333
Gi1/2	1,20,100,200,300,333
Gi1/3	1,20,100,200,300,333

Port Vlan in spanning tree forwarding state and not pruned

Gi0/0 1,20,100,200,300,333
Gi0/1 1,20,100,200,300,333
Gi0/2 1,20,100,200,300,333
Gi0/3 1,100,200,300,333
Gi1/0 1,20,100,200,300,333
Gi1/1 20,100,200,300,333
Gi1/2 1,20,100,200,300,333
Gi1/3 1,20,100,200,300,333
Layer2Switch-2>

Layer2Switch-1>enable
Layer2Switch-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Layer2Switch-1(config)#interface GigabitEthernet0/0
Layer2Switch-1(config-if)#switchport trunk encapsulation dot1q
Layer2Switch-1(config-if)#switchport trunk allowed vlan 20
Layer2Switch-1(config-if)#switchport mode trunk
Layer2Switch-1(config-if)#exit
Layer2Switch-1(config)#interface GigabitEthernet0/1
Layer2Switch-1(config-if)#switchport trunk encapsulation dot1q
Layer2Switch-1(config-if)#switchport trunk native vlan 333
Layer2Switch-1(config-if)#switchport trunk allowed vlan 333
Layer2Switch-1(config-if)#switchport mode trunk
Layer2Switch-1(config-if)#exit
Layer2Switch-1(config)#end
Layer2Switch-1#

```
Layer2Switch-2>enable
Layer2Switch-2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Layer2Switch-2(config)#interface GigabitEthernet0/0
Layer2Switch-2(config-if)#switchport trunk encapsulation dot1q
Layer2Switch-2(config-if)#switchport trunk allowed vlan 20
Layer2Switch-2(config-if)#switchport mode trunk
Layer2Switch-2(config-if)#exit
Layer2Switch-2(config)#interface GigabitEthernet0/1
Layer2Switch-2(config-if)#switchport trunk encapsulation dot1q
Layer2Switch-2(config-if)#switchport trunk native vlan 333
Layer2Switch-2(config-if)#switchport trunk allowed vlan 333
Layer2Switch-2(config-if)#switchport mode trunk
Layer2Switch-2(config-if)#exit
```

```
Layer2Switch-2(config)#interface GigabitEthernet0/1
Layer2Switch-2(config-if)#switchport trunk encapsulation dot1q
Layer2Switch-2(config-if)#switchport trunk native vlan 333
Layer2Switch-2(config-if)#switchport trunk allowed vlan 333
Layer2Switch-2(config-if)#switchport mode trunk
Layer2Switch-2(config-if)#exit
Layer2Switch-2(config)#interface GigabitEthernet0/2
Layer2Switch-2(config-if)#switchport trunk encapsulation dot1q
Layer2Switch-2(config-if)#switchport trunk allowed vlan 20
Layer2Switch-2(config-if)#switchport mode trunk
Layer2Switch-2(config-if)#exit
Layer2Switch-2(config)#interface GigabitEthernet0/3
```

```
Layer2Switch-2(config-if)#switchport trunk encapsulation dot1q
Layer2Switch-2(config-if)#switchport trunk native vlan 333
Layer2Switch-2(config-if)#switchport trunk allowed vlan 333
Layer2Switch-2(config-if)#switchport mode trunk
```

```
Layer2Switch-2(config)#interface GigabitEthernet1/0
Layer2Switch-2(config-if)#switchport trunk encapsulation dot1q
Layer2Switch-2(config-if)#switchport trunk allowed vlan 20
Layer2Switch-2(config-if)#switchport mode trunk
Layer2Switch-2(config-if)#exit
Layer2Switch-2(config)#interface GigabitEthernet1/1
Layer2Switch-2(config-if)#switchport trunk encapsulation dot1q
Layer2Switch-2(config-if)#switchport trunk native vlan 333
Layer2Switch-2(config-if)#switchport trunk allowed vlan 333
Layer2Switch-2(config-if)#switchport mode trunk
Layer2Switch-2(config-if)#exit
Layer2Switch-2(config)#
```

```
Layer2Switch-2(config)#interface GigabitEthernet1/2
Layer2Switch-2(config-if)#switchport trunk encapsulation dot1q
Layer2Switch-2(config-if)#switchport trunk allowed vlan 20
Layer2Switch-2(config-if)#switchport mode trunk
Layer2Switch-2(config-if)#
Layer2Switch-2(config-if)#exit
Layer2Switch-2(config)#interface GigabitEthernet1/3
Layer2Switch-2(config-if)#switchport trunk encapsulation dot1q
Layer2Switch-2(config-if)#switchport trunk native vlan 333
Layer2Switch-2(config-if)#switchport trunk allowed vlan 333
```

```
Layer2Switch-2(config-if)#switchport mode trunk
```

```
Layer2Switch-2(config-if)#exit
```

```
Layer2Switch-2(config)#end
```

```
Layer2Switch-2#
```

```
*Sep  8 11:59:50.015: %SYS-5-CONFIG_I: Configured from console by console
```

```
Layer2Switch-2#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	1
Gi0/1	on	802.1q	trunking	333
Gi0/2	on	802.1q	trunking	1
Gi0/3	on	802.1q	trunking	333
Gi1/0	on	802.1q	trunking	1
Gi1/1	on	802.1q	trunking	333
Gi1/2	on	802.1q	trunking	1
Gi1/3	on	802.1q	trunking	333

Port	Vlans allowed on trunk
Gi0/0	20
Gi0/1	333
Gi0/2	20
Gi0/3	333
Gi1/0	20
Gi1/1	333
Gi1/2	20
Gi1/3	333

Port	Vlans allowed and active in management domain
Gi0/0	20

Port	Vlans allowed and active in management domain
Gi0/1	333
Gi0/2	20
Gi0/3	333
Gi1/0	20
Gi1/1	333
Gi1/2	20
Gi1/3	333

Port	Vlans in spanning tree forwarding state and not pruned
Gi0/0	20
Gi0/1	333
Gi0/2	20
Gi0/3	333
Gi1/0	20
Gi1/1	333
Gi1/2	20
Gi1/3	none

Layer2Switch-2#

Layer2Switch-5#enable

Layer2Switch-5#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Layer2Switch-5(config)#interface GigabitEthernet0/2

Layer2Switch-5(config-if)#switchport trunk encapsulation dot1q


```

Layer2Switch-5(config-if)#switchport trunk allowed vlan 20
Layer2Switch-5(config-if)#switchport mode trunk
Layer2Switch-5(config-if)#exit
Layer2Switch-5(config)#interface GigabitEthernet0/3
Layer2Switch-5(config-if)#switchport trunk encapsulation dot1q
Layer2Switch-5(config-if)#switchport trunk native vlan 333
Layer2Switch-5(config-if)#switchport trunk allowed vlan 333
Layer2Switch-5(config-if)#switchport mode trunk
Layer2Switch-5(config-if)#exit
Layer2Switch-5(config)#end
Layer2Switch-5#

```

*Sep 8 12:31:58.963: %SYS-5-CONFIG_I: Configured from console by console

```

Layer2Switch-5#show interfaces trunk

```

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	1
Gi0/1	on	802.1q	trunking	333
Gi0/2	on	802.1q	trunking	1
Gi0/3	on	802.1q	trunking	333

Port	Vlans allowed on trunk
Gi0/0	20
Gi0/1	333
Gi0/2	20
Gi0/3	333

Port	Vlans allowed and active in management domain
Gi0/0	20
Gi0/1	333

Gi0/2 20
Gi0/3 333

Port Vlans in spanning tree forwarding state and not pruned
Gi0/0 none
Gi0/1 none
Gi0/2 20

Port Vlans in spanning tree forwarding state and not pruned
Gi0/3 333
Layer2Switch-5#

```
Layer2Switch-4>enable
Layer2Switch-4#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Layer2Switch-4(config)#interface GigabitEthernet0/2
Layer2Switch-4(config-if)#switchport trunk encapsulation dot1q
Layer2Switch-4(config-if)#switchport trunk allowed vlan 20
Layer2Switch-4(config-if)#switchport mode trunk
Layer2Switch-4(config-if)#exit
Layer2Switch-4(config)#interface GigabitEthernet0/3
Layer2Switch-4(config-if)#switchport trunk encapsulation dot1q
Layer2Switch-4(config-if)#switchport trunk native vlan 333
Layer2Switch-4(config-if)#switchport trunk allowed vlan 333
Layer2Switch-4(config-if)#switchport mode trunk
Layer2Switch-4(config-if)#exit
Layer2Switch-4(config)#end
Layer2Switch-4#
```

*Sep 8 12:37:37.081: %SYS-5-CONFIG_I: Configured from console by console

Layer2Switch-4#show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	1
Gi0/1	on	802.1q	trunking	333
Gi0/2	on	802.1q	trunking	1
Gi0/3	on	802.1q	trunking	333

Port Vlan allowed on trunk

Gi0/0	20
Gi0/1	333
Gi0/2	20
Gi0/3	333

Port Vlan allowed and active in management domain

Gi0/0	20
Gi0/1	333
Gi0/2	20
Gi0/3	333

Port Vlan in spanning tree forwarding state and not pruned

Gi0/0	20
Gi0/1	333
Gi0/2	20

Port Vlan in spanning tree forwarding state and not pruned

Gi0/3 333

Layer2Switch-4#

Layer2Switch-1#show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	1
Gi0/1	on	802.1q	trunking	333
Gi0/2	on	802.1q	trunking	1
Gi0/3	on	802.1q	trunking	333
Gi1/0	on	802.1q	trunking	1
Gi1/1	on	802.1q	trunking	333
Gi1/2	on	802.1q	trunking	1
Gi1/3	on	802.1q	trunking	333

Port Vlan allowed on trunk

Gi0/0	20
Gi0/1	333
Gi0/2	20
Gi0/3	333
Gi1/0	20
Gi1/1	333
Gi1/2	20
Gi1/3	333

Port Vlan allowed and active in management domain

Gi0/0	20
-------	----

Port Vlan allowed and active in management domain

Gi0/1	333
Gi0/2	20
Gi0/3	333
Gi1/0	20
Gi1/1	333
Gi1/2	20
Gi1/3	333

Port	Vlans in spanning tree forwarding state and not pruned
Gi0/0	none
Gi0/1	none
Gi0/2	20
Gi0/3	333
Gi1/0	20
Gi1/1	333
Gi1/2	20
Gi1/3	333

Layer2Switch-1#

Layer2Switch-3#show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	1
Gi0/1	on	802.1q	trunking	333
Gi0/2	on	802.1q	trunking	1
Gi0/3	on	802.1q	trunking	333

Port	Vlans allowed on trunk
Gi0/0	20

Gi0/1	333
Gi0/2	20
Gi0/3	333

Port	Vlans allowed and active in management domain
Gi0/0	20
Gi0/1	333
Gi0/2	20
Gi0/3	333

Port	Vlans in spanning tree forwarding state and not pruned
Gi0/0	none
Gi0/1	none
Gi0/2	20

Port	Vlans in spanning tree forwarding state and not pruned
Gi0/3	333

Layer2Switch-3#

Layer2Switch-4#show vlan brief

VLAN Name	Status	Ports
1 default	active	
20 VLAN20	active	Gi1/0
100 VLAN100	active	
200 VLAN0200	active	
300 VLAN0300	active	
333 VLAN333	active	Gi1/1

```

1002 fddi-default          act/unsup
1003 trcrf-default         act/unsup
1004 fddinet-default       act/unsup
1005 trbrf-default        act/unsup
Layer2Switch-4#

```

Layer2Switch-5#show vlan brief

VLAN Name	Status	Ports

1 default	active	
20 VLAN20	active	Gi1/0
100 VLAN100	active	
200 VLAN0200	active	
300 VLAN0300	active	
333 VLAN333	active	Gi1/1
1002 fddi-default	act/unsup	
1003 trcrf-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trbrf-default	act/unsup	

Layer2Switch-5#

ЗАДАНИЕ 2

2) Проверить доступность персональных компьютеров, находящихся в одинаковых VLAN и недоступность находящихся в различных, результаты задокументировать

```
PC1> ip 192.168.10.1/24
```

```
Checking for duplicate address...
```

```
PC1 : 192.168.10.1 255.255.255.0
```

```
PC1> save
```

```
Saving startup configuration to startup.vpc
```

```
. done
```

```
PC2> ip 192.168.10.2/24
```

```
Checking for duplicate address...
```

```
PC2 : 192.168.10.2 255.255.255.0
```

```
PC2> save
```

```
Saving startup configuration to startup.vpc
```

```
. done
```

```
PC4> ip 192.168.10.4/24
```

```
Checking for duplicate address...
```

```
PC4 : 192.168.10.4 255.255.255.0
```

```
PC4> save
```

```
Saving startup configuration to startup.vpc
```

```
. done
```

```
PC5> ip 192.168.10.5/24
```

```
Checking for duplicate address...
```

```
PC5 : 192.168.10.5 255.255.255.0
```


PC5> save

Saving startup configuration to startup.vpc

. done

PC6> ip 192.168.10.6/24

Checking for duplicate address...

PC6 : 192.168.10.6 255.255.255.0

PC6> save

Saving startup configuration to startup.vpc

. done

PC1> ping 192.168.10.3

84 bytes from 192.168.10.3 icmp_seq=1 ttl=64 time=69.240 ms

84 bytes from 192.168.10.3 icmp_seq=2 ttl=64 time=101.276 ms

84 bytes from 192.168.10.3 icmp_seq=3 ttl=64 time=45.062 ms

84 bytes from 192.168.10.3 icmp_seq=4 ttl=64 time=75.448 ms

84 bytes from 192.168.10.3 icmp_seq=5 ttl=64 time=71.646 ms

PC1> ping 192.168.10.5

84 bytes from 192.168.10.5 icmp_seq=1 ttl=64 time=60.173 ms

84 bytes from 192.168.10.5 icmp_seq=2 ttl=64 time=30.594 ms

84 bytes from 192.168.10.5 icmp_seq=3 ttl=64 time=52.932 ms

84 bytes from 192.168.10.5 icmp_seq=4 ttl=64 time=66.946 ms

84 bytes from 192.168.10.5 icmp_seq=5 ttl=64 time=51.535 ms

PC3> ping 192.168.10.5

84 bytes from 192.168.10.5 icmp_seq=1 ttl=64 time=86.946 ms

84 bytes from 192.168.10.5 icmp_seq=2 ttl=64 time=31.569 ms

84 bytes from 192.168.10.5 icmp_seq=3 ttl=64 time=61.949 ms
84 bytes from 192.168.10.5 icmp_seq=4 ttl=64 time=31.677 ms
84 bytes from 192.168.10.5 icmp_seq=5 ttl=64 time=65.452 ms

PC2> ping 192.168.10.4

84 bytes from 192.168.10.4 icmp_seq=1 ttl=64 time=74.546 ms
84 bytes from 192.168.10.4 icmp_seq=2 ttl=64 time=35.311 ms
84 bytes from 192.168.10.4 icmp_seq=3 ttl=64 time=55.466 ms
84 bytes from 192.168.10.4 icmp_seq=4 ttl=64 time=38.603 ms
84 bytes from 192.168.10.4 icmp_seq=5 ttl=64 time=43.534 ms

PC2> ping 192.168.10.6

84 bytes from 192.168.10.6 icmp_seq=1 ttl=64 time=44.776 ms
84 bytes from 192.168.10.6 icmp_seq=2 ttl=64 time=67.447 ms
84 bytes from 192.168.10.6 icmp_seq=3 ttl=64 time=82.440 ms
84 bytes from 192.168.10.6 icmp_seq=4 ttl=64 time=51.399 ms
84 bytes from 192.168.10.6 icmp_seq=5 ttl=64 time=63.321 ms

PC4> ping 192.168.10.6

84 bytes from 192.168.10.6 icmp_seq=1 ttl=64 time=37.291 ms
84 bytes from 192.168.10.6 icmp_seq=2 ttl=64 time=62.967 ms
84 bytes from 192.168.10.6 icmp_seq=3 ttl=64 time=52.339 ms
84 bytes from 192.168.10.6 icmp_seq=4 ttl=64 time=93.552 ms
84 bytes from 192.168.10.6 icmp_seq=5 ttl=64 time=53.003 ms

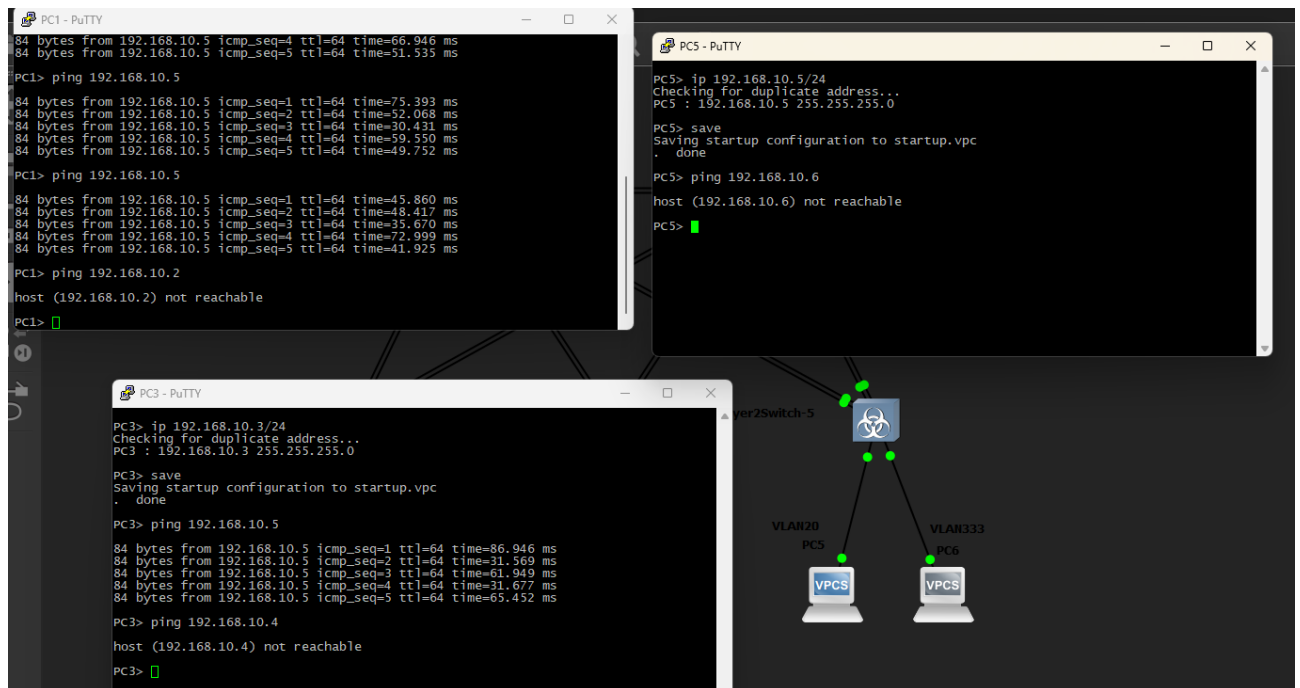


Рисунок 2 — Проверка доступности узлов в одинаковых VLAN и недоступности узлов в разных VLAN

PC1> ping 192.168.10.2

host (192.168.10.2) not reachable

PC3> ping 192.168.10.4

host (192.168.10.4) not reachable

PC5> ping 192.168.10.6

host (192.168.10.6) not reachable

ЗАДАНИЕ 3

3) Перехватить в Wireshark пакеты с тегами и без тегов (nb!), результаты задокументировать

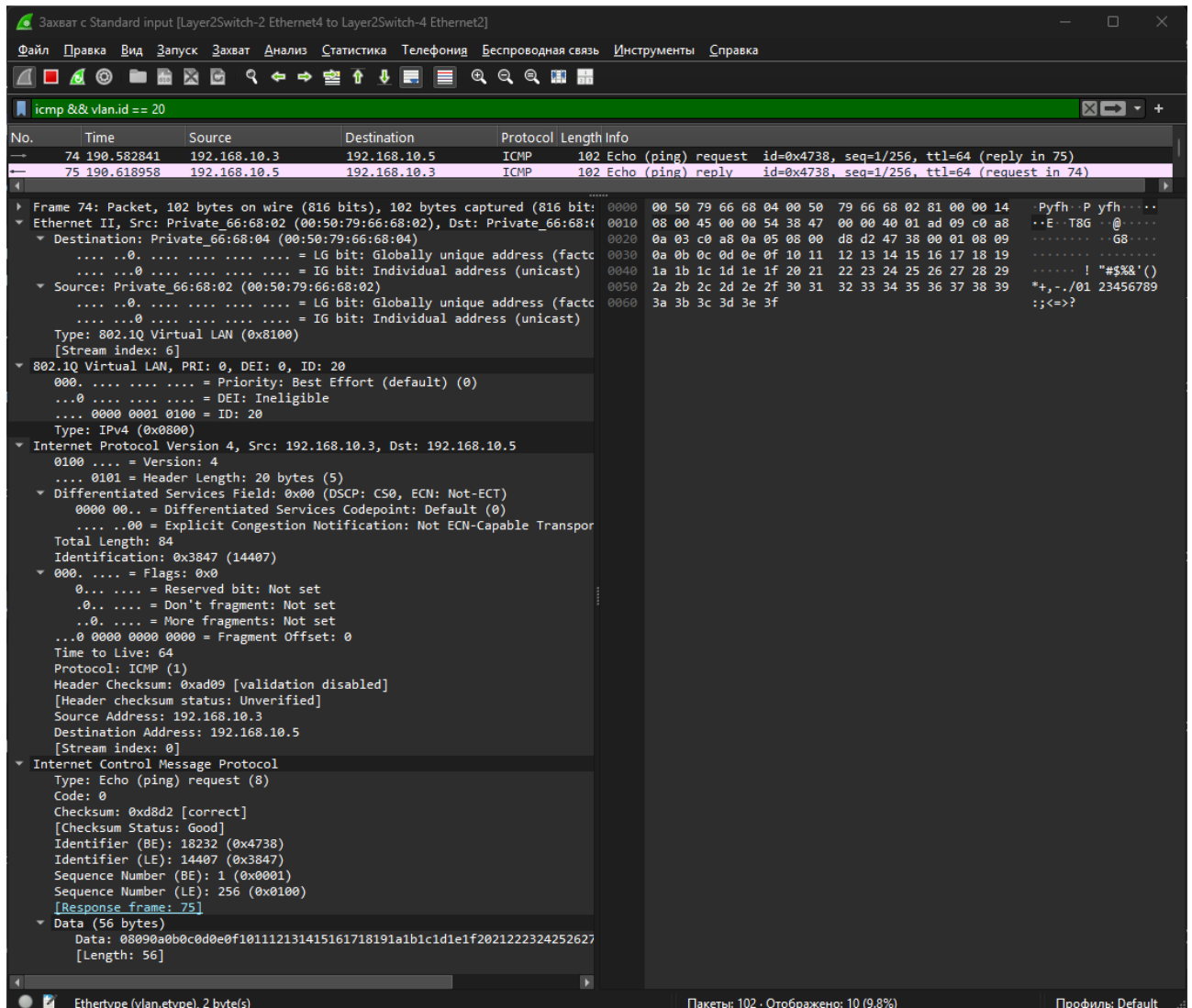


Рисунок 3 — Перехват ICMP-пакета VLAN20 с тегом IEEE 802.1Q

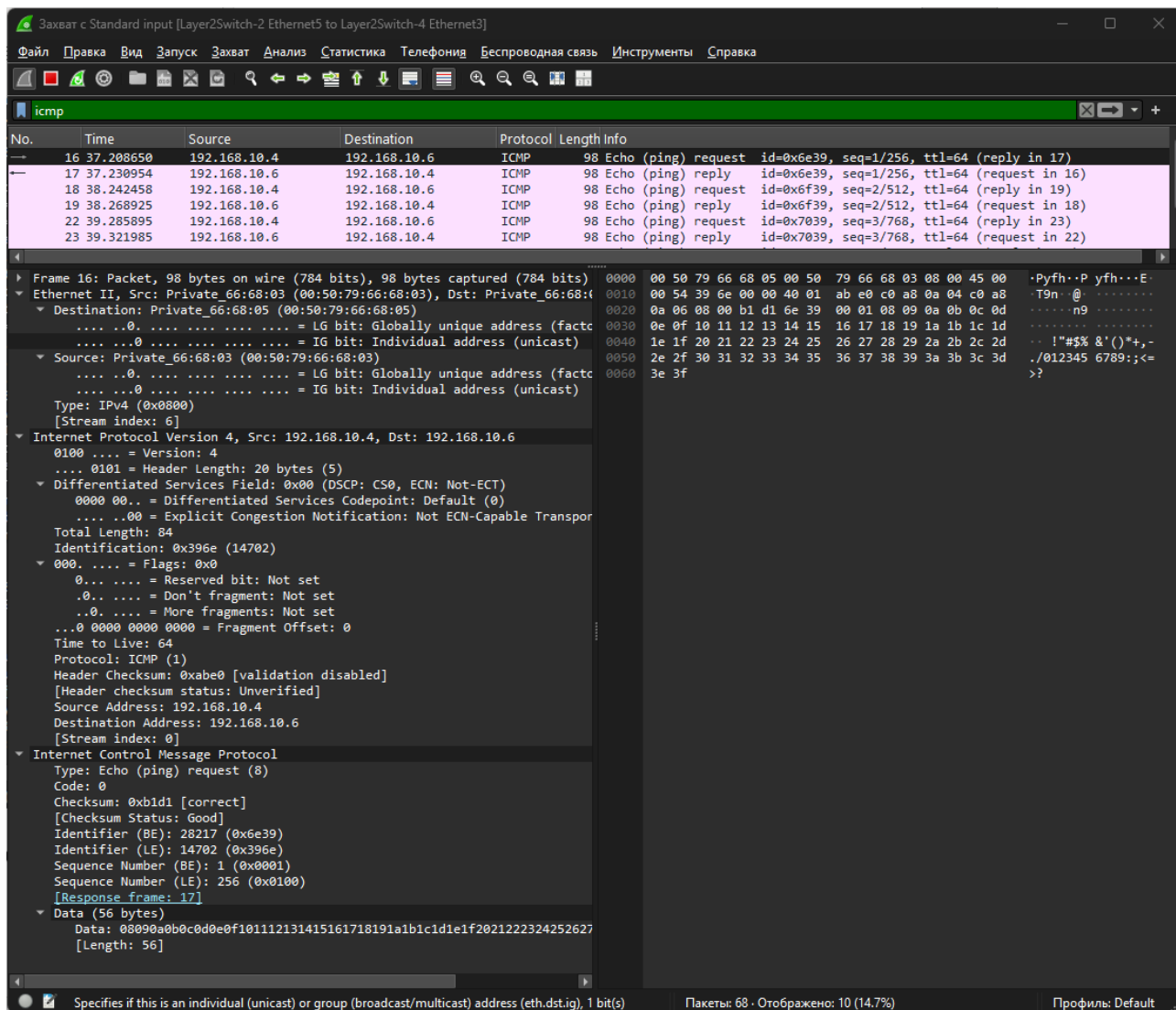


Рисунок 4 — Перехват ICMP-пакета Native VLAN333 без тега IEEE 802.1Q

ЗАДАНИЕ 4

4) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств

```
Layer2Switch-1>enable
```

```
Layer2Switch-1#copy running-config startup-config
```

```
Destination filename [startup-config]?
```

```
Building configuration...
```

```
Compressed configuration from 6107 bytes to 2313 bytes[OK]
```

```
Layer2Switch-1#
```

```
*Sep  8 12:26:08.244: %GRUB-5-CONFIG_WRITING: GRUB configuration  
is being updated on disk. Please wait...
```

```
*Sep  8 12:26:10.052: %GRUB-5-CONFIG_WRITTEN: GRUB configuration  
was written to disk successfully.
```

```
Layer2Switch-1#
```

```
Layer2Switch-2>enable
```

```
Layer2Switch-2#copy running-config startup-config
```

```
Destination filename [startup-config]?
```

```
Building configuration...
```

```
Compressed configuration from 6100 bytes to 2309 bytes[OK]
```

```
*Sep  8 12:19:52.592: %GRUB-5-CONFIG_WRITING: GRUB configuration  
is being updated on disk. Please wait...
```

```
Layer2Switch-2#
```

```
*Sep  8 12:19:54.464: %GRUB-5-CONFIG_WRITTEN: GRUB configuration  
was written to disk successfully.
```

```
Layer2Switch-2#
```

```
Layer2Switch-3>enable
```

```
Layer2Switch-3#copy running-config startup-config
```

```
Destination filename [startup-config]?
```

```
Building configuration...
```

Compressed configuration from 5514 bytes to 2140 bytes[OK]

*Sep 8 12:57:30.881: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...

Layer2Switch-3#

*Sep 8 12:57:33.037: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.

Layer2Switch-3#

Layer2Switch-4>enable

Layer2Switch-4#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

Compressed configuration from 5514 bytes to 2141 bytes[OK]

Layer2Switch-4#

*Sep 8 12:59:38.479: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...

*Sep 8 12:59:40.268: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.

Layer2Switch-4#

Layer2Switch-5>enable

Layer2Switch-5#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

Compressed configuration from 5514 bytes to 2143 bytes[OK]

*Sep 8 12:55:12.413: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...

Layer2Switch-5#

*Sep 8 12:55:14.391: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.

Layer2Switch-5#

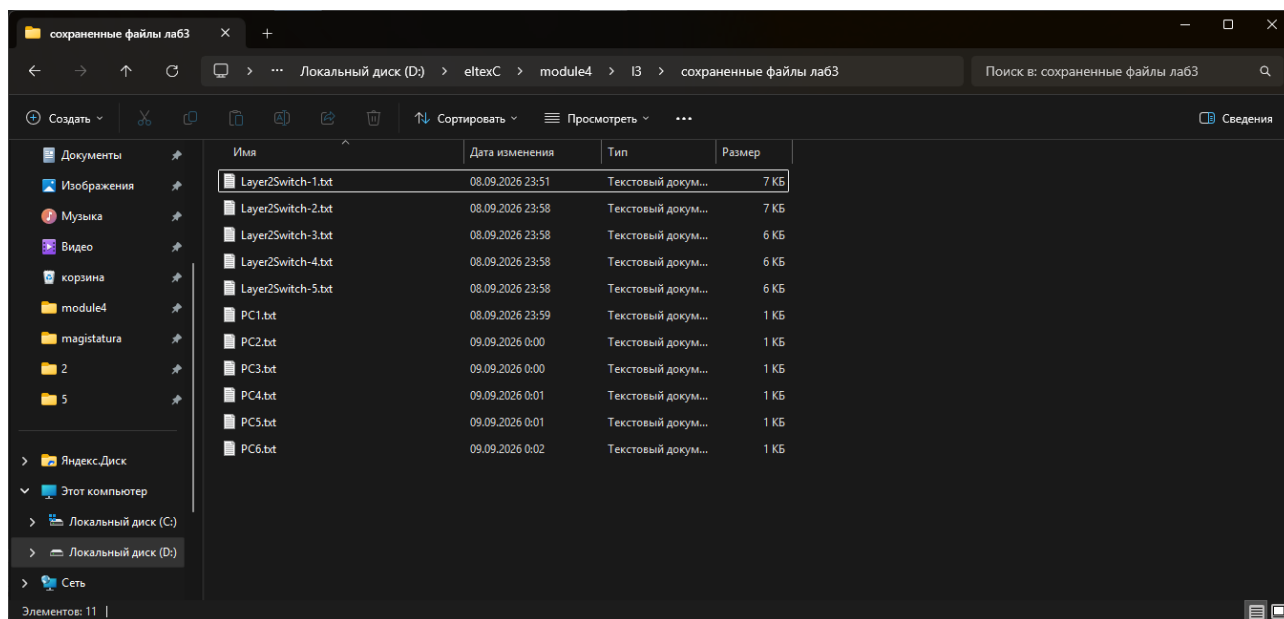


Рисунок 5 — Набор файлов конфигураций устройств